Que 1: What is Data Science? Explain its primary objective.

Data Science is an interdisciplinary field that uses statistics, mathematics, programming, machine learning, and domain knowledge to extract useful insights from data.

Primary Objective:

- To convert raw data into meaningful information.
- To support better decision-making.
- To predict future outcomes.
- To solve real-world problems using data.

Que 2: Why is Data Science important in today's digital world? Mention any four reasons.

Importance of Data Science:

- Helps organizations make data-driven decisions.
- Improves business efficiency and productivity.
- Predicts future trends using machine learning.
- Enables personalization in applications like Netflix and Amazon.

Que 3: List any four benefits of Data Science in business and society.

Benefits:

- Better decision-making.
- Cost reduction.
- Fraud detection.
- Improved healthcare through disease prediction.

Que 4: Mention four real-world applications of Data Science.

Applications:

- Healthcare (disease prediction)
- Banking (fraud detection)
- E-commerce (recommendation systems)
- Transportation (traffic prediction)

Que 5: What are the different facets of data? Briefly explain each.

Facets of Data:

- **Structured Data:** Organized in rows and columns (Excel, SQL database).
- **Semi-Structured Data:** Contains tags but no fixed format (XML, JSON).
- **Unstructured Data:** No predefined structure (images, videos, emails).

Que 6: Differentiate between structured, semi-structured, and unstructured data with suitable examples.

Structured	Semi-Structured	Unstructured
Fixed format	Partial structure	No structure
Stored in databases	XML, JSON	Images, audio, video
Easy to analyze	Moderately easy	Difficult to analyze
Example: Student database	Example: XML file	Example: YouTube video

Que 7: What is Big Data? State its key characteristics (5 Vs).

Big Data refers to extremely large and complex datasets that cannot be processed using traditional methods.

5 Vs:

- **Volume** – Huge amount of data
- **Velocity** – Speed of data generation
- **Variety** – Different data formats
- **Veracity** – Data quality and reliability
- **Value** – Useful insights from data

Que 8: Explain the Big Data ecosystem. Name any four technologies used in it.

The **Big Data ecosystem** is a collection of tools used to store, process, analyze, and visualize large datasets.

Technologies:

- Hadoop
- Spark
- Hive
- HBase

Que 9: What are the major phases of the Data Science process?

- Data Collection
- Data Cleaning
- Data Integration
- Data Transformation
- Data Analysis
- Model Building
- Model Evaluation
- Deployment and Presentation

Que 10: What is data retrieval? Mention two common sources from which data can be retrieved.

Data Retrieval is the process of collecting data from different sources.

Sources:

- Databases
- Websites/APIs

Que 11: Why is data collection considered an important step in the Data Science process?

Data collection is important because:

- It provides quality data for analysis.
- Improves model accuracy.
- Reduces errors.

- Ensures reliable decision-making.

Que 12: What is data cleansing? List any four common data-cleaning techniques.

Data Cleansing is the process of correcting or removing inaccurate and inconsistent data.

Techniques:

- Remove duplicates
- Handle missing values
- Correct inconsistent data
- Remove outliers

Que 13: What is data integration? Why is it necessary in Data Science?

Data Integration is combining data from multiple sources into one dataset.

Need:

- Provides complete information.
- Improves data consistency.
- Removes redundancy.
- Supports better analysis.

Que 14: Explain data transformation with suitable examples.

Data Transformation converts data into a suitable format for analysis.

Examples:

- Converting text into numerical values.
- Normalizing data.
- Scaling numerical values.
- Date format conversion.

Que 15: What is data analysis? Mention any four techniques used for data analysis.

Data Analysis is the process of examining data to discover useful information.

Techniques:

- Statistical Analysis

- Data Mining
- Machine Learning
- Visualization

Que 16: Differentiate between descriptive, predictive, and prescriptive analytics.

Descriptive	**Predictive**	**Prescriptive**
What happened?	What will happen?	What should be done?
Uses historical data	Uses ML models	Suggests actions
Reports and dashboards	Forecasting	Decision optimization

Que 17: What is meant by building a model in Data Science? Mention the basic steps involved.

Model Building is creating a machine learning model that learns patterns from data.

Steps:

1. Select algorithm
2. Split data
3. Train model
4. Test model
5. Improve performance

Que 18: Why is model evaluation important? Name any four evaluation metrics used for classification models.

Model evaluation measures how well the model performs.

Metrics:

- Accuracy
- Precision
- Recall
- F1-Score

Que 19: Why is presenting findings an essential part of the Data Science process? Mention any four visualization tools or techniques.

Presenting findings helps stakeholders understand insights and make informed decisions.

Visualization tools/techniques:

- Bar Chart
- Pie Chart
- Line Graph
- Tableau (or Power BI)

Que 20: How are Data Science models deployed into real-world applications? Give any four examples of Data Science applications.

After testing, models are deployed as **web applications, mobile apps, cloud services, or APIs** so users can access predictions in real time.

Applications:

- Spam email detection
- Disease prediction
- Movie recommendation systems
- Credit card fraud detection